12.4.7 Marginal Revenue

Marginal Revenue is the revenue earned by selling an additional unit of the product. In other words, Marginal Revenue is an addition made to the total revenue by selling one more unit of the good.

$$MR = \frac{d(TR)}{dQ} \; or = \frac{\Delta TR}{\Delta Q}$$

Where ΔTR stands for change in the total revenue, and ΔQ stands for change in output.

Example: 12.18

Given TR = $50Q - 4Q^2$,find marginal revenue when Q = 3.

Solution:

$TR = 50Q - 4Q^2$

$MR = d(TR)/dQ$

$MR = 50(1)Q^{1-1} - 4(2)Q^{2-1}$

$= 50(1)Q^0 - 8\,Q^1$

$= 50(1) - 8Q \quad (\therefore Q^0 = 1, Q^1 = Q)$

$MR = 50 - 8Q$

When Q = 3

$MR = 50 - 8(3) = 26$

Example: 12.19

A producer has the total cost function $TC(Q) = Q^3 - 18Q^2 + 91Q + 10$ where costs are given in rupees. Find the marginal cost (MC) and the average variable cost (AVC), when Q =3.

Solution:

Given $TC(Q) = Q^3 - 18Q^2 + 91Q + 10$, To find MC differentiate the function with respect to Q.

$$MC(Q) = \frac{d(TC)}{dQ} = 3Q^{3-1} - 18(2)Q^{2-1} + 91(1)Q^{1-1} + 0$$

$= 3Q^2 - 36Q^1 + 91Q^0 + 0$

$MC(Q) = 3Q^2 - 36Q + 91 \quad (\therefore Q^0 = 1)$

When Q = 3

$MC(Q) = 3(3^2) - 36(3) + 91$

$= 3(9) - 108 + 91$

$= 27 - 108 + 91$

$= 118 - 108$

$= 10$

To find AVC

Given $TC(Q) = Q^3 - 18Q^2 + 91Q + 10$

We know TVC (Q) $= Q^3 - 18Q^2 + 91Q$ ($\therefore$constant value is fixed cost)

AVC(Q)=TVC(Q)/Q

$$\boxed{AVC(Q) = Q^2 - 18Q + 91}$$

When Q = 3

$AVC(Q) = 3^2 - 18(3) + 9$

$= 9 - 54 + 91$

$= 100 - 54 = 46$

$$\therefore AVC(Q) = \frac{Q^3 - 18Q^2 + 91Q}{Q}$$

$= Q^2 - 18Q + 91$

Note : Fixed cost= 10

Average fixed cost = $\frac{10}{Q}$.

$$\text{Average cost} = \frac{Q^3 - 18Q^2 + 91Q + 10}{Q}$$

$$= Q^2 - 18Q + 91 + \frac{10}{Q}$$

So Average cost = AVC + AFC

AC = AVC + AFC

$$AC = \frac{TC}{Q}$$

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